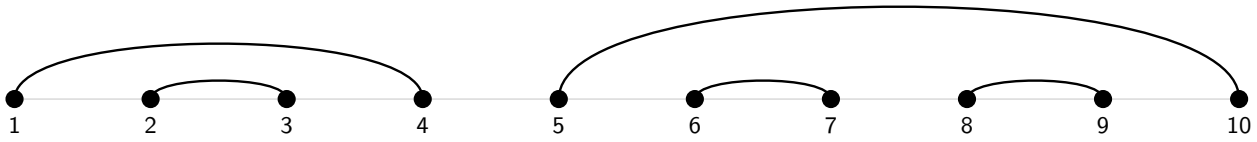
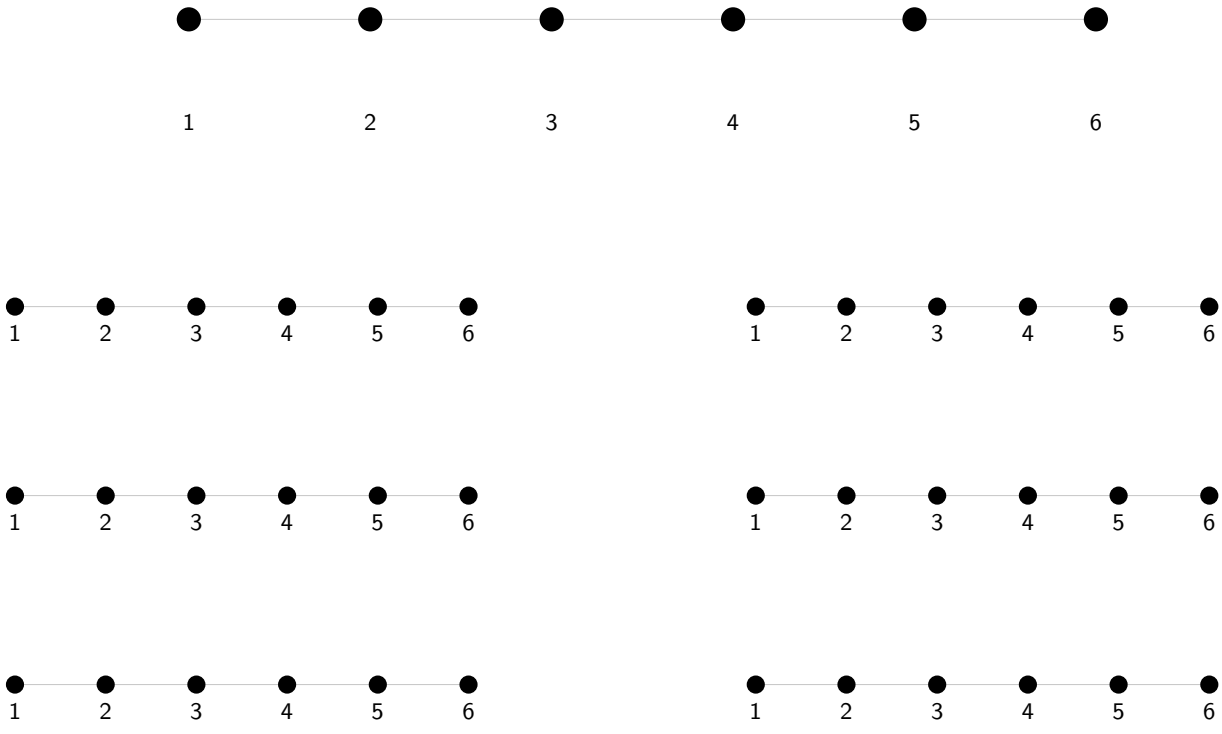
Week 12 / Catalan bijections / Grades 2–3

Use each dot once. Keep strings above the row without crossings. Keep dots in place. Pairings are different when a dot has a different partner.

Example with ten dots:

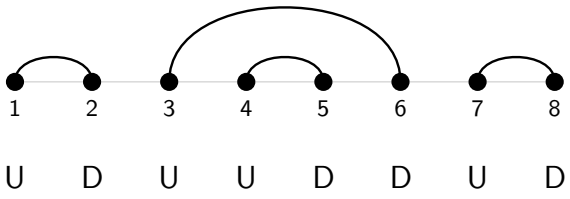


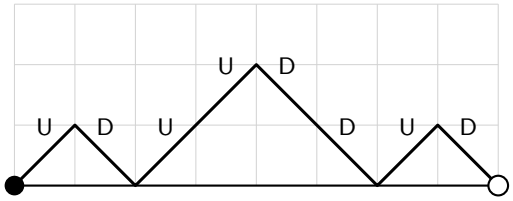
Problem 1: Find every way to pair the six dots. Draw a complete collection with no repeats.



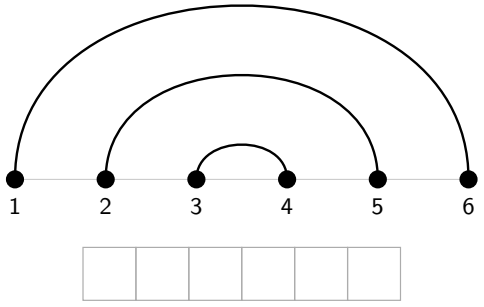
Read the dots from left to right. Write U at the first end of each pair and D at the other end. U is one step up and right; D is one step down and right.

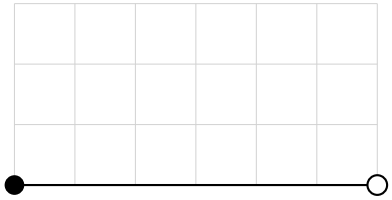
Example:

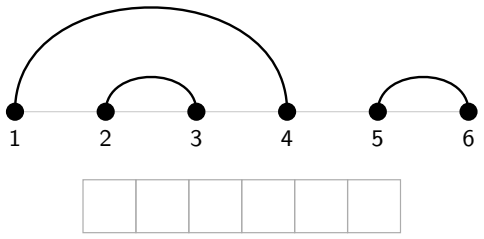


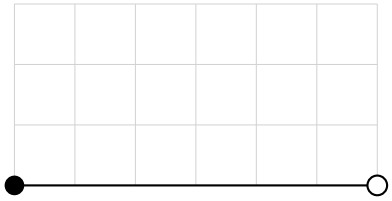


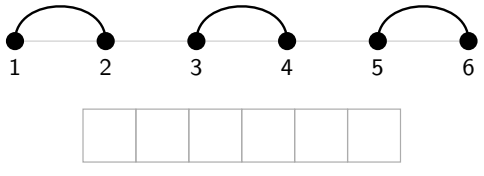
Problem 2: Draw the path for each pairing, starting at the marked dot. Can a pairing ever make a path that goes below the dark line?

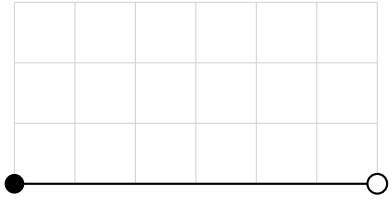




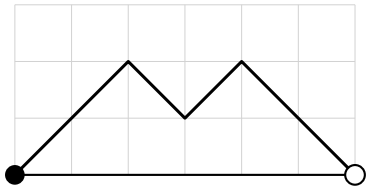




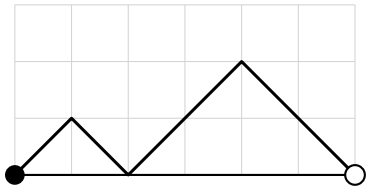




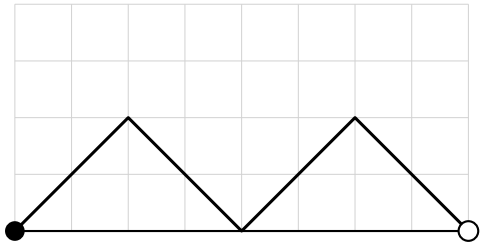
Problem 3: Draw a pairing for each path. Could a path come from two different pairings without crossings? Explain.



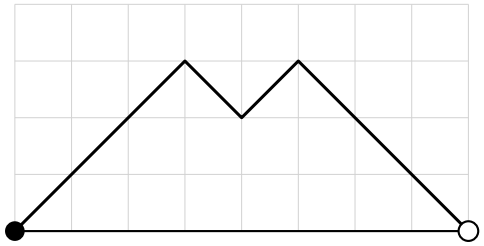






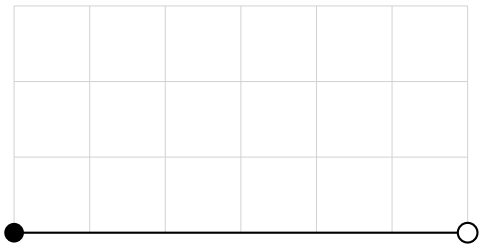


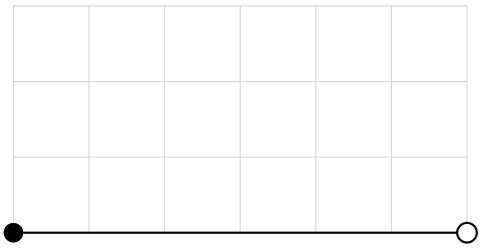


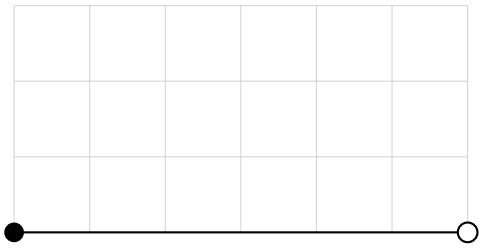


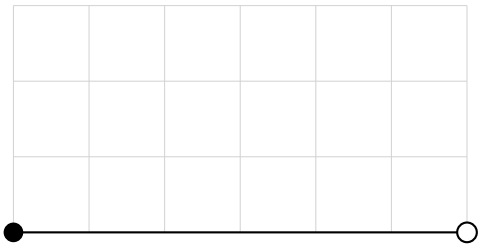


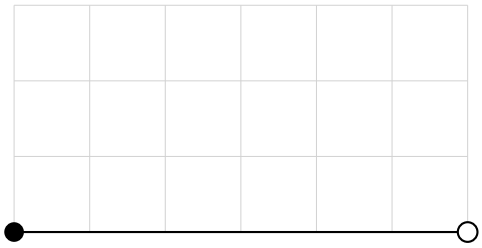
Problem 4: Find every path with three up steps and three down steps that starts and ends on the dark line and never goes below it. Match your paths to your six-dot pairings.

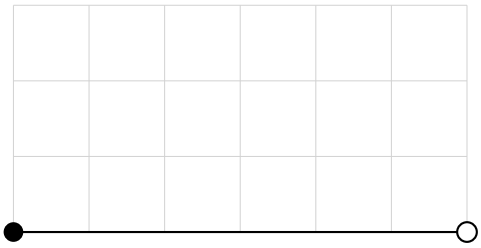












Problem 5: Which strings of letters can be codes for pairings of eight dots? Draw the pairing whenever it is possible, and explain what goes wrong whenever it is impossible.

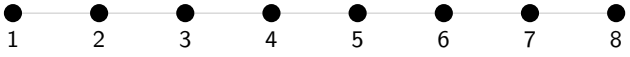
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